1 **(a)** 48 **[B1]**

(b) 56 **[B1]**

(c) $180 - 48 - 56 = 76$ **[B1]**

[Total Marks: 3]

2 **(a)** 125 **[B1]**

(b) Corresponding angles (are equal) **[B1]**

(c) 144 **[B1]**

[Total Marks: 3]

3 **(a)** Identifies alternate angles by drawing a parallel line through E OR forms a triangle by extending a line **[M1]**

85 **[A1]**

(b) $\frac{360}{24}$ **[M1]**

15 **[A1]**

[Total Marks: 4]

4 **(a)** Angle $PRC = 54°$ **[B1]**

Reason: alternate angles (AB is parallel to CD) **[B1]**

(b) Angle $PQR = 180° - 112° = 68°$ **[M1]**

Angle $PRQ = 180° - 54° - 68° = 58$ **[A1]**

(c) 58 **[B1]**

[Total Marks: 5]

5 **(a)** 52 **[B1]**

(b) Angles on a straight line sum to 180°: $180 - (52 + 61)$ **[M1]**

67 **[A1]**

(c) Identifies angle $TVU = 61°$ (alternate angle to QTV) or angle $VTW + TVW$ exterior relationship **[M1]**

Angle $TVW = 180 - 61 = 119$ **[A1]**

[Total Marks: 5]

6 **(a)** 118 **[B1]**

Alternate angles (are equal) **[B1]**

(b) $180° - 118° = 62$ **[B1]**

(c) Uses isosceles triangle GHK: $\angle HGK = 62°$ and $\angle BGH = 180° - 118° = 62°$ (or $\angle BGK = \angle GKD = 180° - 56°$) **[M1]**

124 **[A1]**

[Total Marks: 5]

7 (a) $360 \div 24$ **[M1]**

15 **[A1]**

(b) $(15 - 2) \times 180$ **[M1]**

2340 **[A1]**

(c) Exterior angle $= 180 - 165 = 15°$ **[M1]**

$360 \div 15 = 24$ **[A1]**

[Total Marks: 6]

8 (a) $180 - (55 + 70)$ **[M1]**

55 **[A1]**

(b) 55 **[B1]**

(c) Angle $DAE = 55°$ (from part a) AND Angle $ADE = 55°$ **[B1]**

Two angles are equal, so the triangle is isosceles **[B1]**

[Total Marks: 5]

9 (a) Exterior angle $= \dfrac{360°}{5} = 72°$ or interior sum $= (5 - 2) \times 180° = 540°$ **[M1]**

108 **[A1]**

(b) Uses angles at point B: $\angle ABC = 360° - 108° - 102°$ **[M1]**

$\angle ABC = 150°$ **[A1]**

Exterior angle of regular n-gon $= 180° - 150° = 30°$ **[M1]**

$n = \dfrac{360°}{30°} = 12$ **[A1]**

[Total Marks: 6]

10 (a) (i) $\dfrac{(6 - 2) \times 180°}{6}$ or $180° - \dfrac{360°}{6}$ **[M1]**

120 **[A1]**

(ii) 108 **[B1]**

(b) Angle $ABG = 360° - 120° - 108° = 132°$ **[M1]**

Triangle ABG is isosceles ($AB = BG$) **[M1]**

$\dfrac{180^\circ - 132^\circ}{2} = 24$ **[A1]**

(c) $(2x + 10) + (3x - 15) + (x + 25) + 100 = 360$ **[M1]**

$6x + 120 = 360$ giving $x = 40$ **[A1]**

Largest angle $= 3(40) - 15 = 105$ **[A1]**

[Total Marks: 9]

11 (a) Equates sum of angles to formula: $3 \times 140 + 165(n - 3) = 180(n - 2)$ **[M1]**

Expands brackets correctly: $420 + 165n - 495 = 180n - 360$ **[M1]**

Rearranges to isolate n: $15n = 285$ (or equivalent) **[M1]**

19 **[A1]**

(b) States $a + b + c = 180$ **[B1]**

Substitutes $a + b = c$ to get $c + c = 180 \implies 2c = 180 \implies c = 90$ completing the proof **[B1]**

(c) (i) Calculates exterior angle of $P = \dfrac{360}{12} = 30^\circ$ so exterior angle of $Q = 15^\circ$ **[M1]**

24 **[A1]**

(ii) Calculates interior angles of P and Q: $180 - 30 = 150^\circ$ and $180 - 15 = 165^\circ$ **[M1]**

$360 - (150 + 165) = 45$ **[A1]**

[Total Marks: 10]

12 (a) Sum of interior angles of a pentagon $= (5 - 2) \times 180^\circ = 540^\circ$ **[M1]**

$x + (x + 20) + (2x - 10) + (2x + 30) + 140 = 540$ **[M1]**

$6x + 180 = 540 \implies 6x = 360 \implies x = 60$ **[A1]**

(b) Substituting $x = 60$ into $(2x + 30)$ or checking all angle sizes $(60^\circ, 80^\circ, 110^\circ, 150^\circ, 140^\circ)$ **[M1]**

150 **[A1]**

[Total Marks: 5]

13 (a) Recognises the sum of the interior angles of a pentagon is 540° **[B1]**

Sets up a correct equation: $(2x + 10) + 100 + 80 + 3x + (x + 50) = 540$ **[M1]**

50 **[A1]**

(b) Sets up a relationship for the angles, e.g., $4E + E = 180$ where E is the exterior angle **[M1]**

$E = 36^\circ$ **[M1]**

10 **[A1]**

[Total Marks: 6]

14 (a) $5x + 10 = 7x - 30$ **[M1]**

$x = 20$ **[A1]**

Corresponding angles (are equal) **[B1]**

(b) Angle $GHD = 7(20) - 30 = 110°$ **[B1]**

Angle $GHK = 180 - 110 = 70°$ (Angles on a straight line) **[B1]**

Angle $GKH = 180 - (70 + 35) = 75°$ (Angles in a triangle) **[B1]**

[Total Marks: 6]

15 (a) (i) Equates the sum of the angles to the interior angle sum of a hexagon (720°):
$x + 2x + x + 30 + 2x + 10 + 3x - 20 + 160 = 720$ **[M1]**

Simplifies the equation to $9x + 180 = 720$ (or $9x = 540$) **[M1]**

60 **[A1]**

(ii) Sets up an equation for the exterior angle e:
$6.5e + e = 180°$ **[M1]**

$7.5e = 180° \implies e = 24°$ **[M1]**

Number of sides $n = \dfrac{360}{24} = 15$ **[A1]**

(iii) Let exterior angles be E_m and E_n. Sets up the first equation:
$E_m + E_n = 54$ **[B1]**

Relates interior difference to exterior difference:
$(180 - E_m) - (180 - E_n) = 18 \implies E_n - E_m = 18$ **[M1]**

Solves the simultaneous equations by addition (or substitution):
$2E_n = 72 \implies E_n = 36°$ **[M1]**

Calculates n: $n = \dfrac{360}{36} = 10$ **[A1]**

Calculates m: $E_m = 18° \implies m = \dfrac{360}{18} = 20$ **[A1]**

[Total Marks: 11]

16 (a) Recognises that $\angle PXQ = \angle BPX + \angle DQX$ (e.g. by drawing a parallel line through X to form alternate angles) **[M1]**

$(3x + 15) + (2x + 10) = 75$ **[M1]**

$5x + 25 = 75 \implies 5x = 50 \implies x = 10$ **[A1]**

(b) Expression for exterior or interior angle of P: $\dfrac{360}{n}$ or $180 - \dfrac{360}{n}$ **[B1]**

Expression for exterior or interior angle of Q: $\dfrac{360}{3n}$ or $180 - \dfrac{120}{n}$ **[B1]**

Forms a correct equation: $\left(180 - \dfrac{120}{n}\right) - \left(180 - \dfrac{360}{n}\right) = 16$ OR $\dfrac{360}{n} - \dfrac{120}{n} = 16$ **[M1]**

$\dfrac{240}{n} = 16 \implies n = 15$ **[A1]**

[Total Marks: 7]

17 **(a)** $\dfrac{(5-2)\times 180}{5}$ OR $180-\dfrac{360}{5}$ **[M1]**

108 **[A1]**

(b) Interior angle of hexagon $= 120°$ **[B1]**

Angle $AEI = 360 - 108 - 120 = 132$ **[A1]**

(c) $AE = EI$ (sides of regular polygons sharing an edge are equal) **[M1]**

Triangle AEI is isosceles **[M1]**

Angle $EAI = \dfrac{180-132}{2} = 24$ **[A1]**

[Total Marks: 7]

18 **(a)** $(5-2)\times 180 \div 5$ OR $180-(360\div 5)$ **[M1]**

108 **[A1]**

(b) Interior angle of square $= 90°$ **[B1]**

Interior angle of n-gon $= 360 - 90 - 108 = 162°$ **[M1]**

$n = 360 \div (180 - 162) = 20$ **[A1]**

(c) Recognises triangle PFA is isosceles (since $PA = PF$ from shared sides of regular polygons) **[M1]**

Angle $PFA = (180 - 162) \div 2 = 9$ **[A1]**

[Total Marks: 7]

19 **(a)** Calculates interior angle sum of a hexagon: $(6-2)\times 180° = 720°$ **[B1]**

Sums algebraic expressions:
$(2x+15)+(3x-20)+(2x+35)+(4x-45)+(x+50)+(3x+10) = 720$ **[M1]**

Simplifies to $15x + 45 = 720$ **[M1]**

45 **[A1]**

(b) Calculates $\angle Q = 3(45) - 20 = 115°$, $\angle R = 2(45) + 35 = 125°$, and $\angle S = 4(45) - 45 = 135°$ **[M1]**

Finds sum of interior angles along QRS: $115° + 125° + 135° = 375°$ (or constructs line through R parallel to PQ, giving co-interior angles $65° + 60° = 125° e 135°$) **[M1]**

States No, because the sum of interior angles $\angle Q + \angle R + \angle S = 375° e 360°$ (or corresponding/alternate angles are not equal) **[A1]**

[Total Marks: 7]

20 **(a)** Finds an expression for the interior angle ABC using the sum of angles at a point:
Interior angle $ABC = 360° - (300 - 2x)° = (60 + 2x)°$ **[M1]**

Draws a parallel line through B and uses alternate angles to form an equation:
Interior angle $ABC = (3x + 10)° + (2x + 20)°$ **[M1]**

Equates the two expressions for the interior angle ABC:

$5x + 30 = 60 + 2x$ **[M1]**

Solves the equation correctly to show the result:
$3x = 30 \implies x = 10$ **[A1]**

(b) Calculates the interior angle ABC using $x = 10$:
Angle $ABC = 60 + 2(10) = 80°$ **[M1]**

Recognises triangle ABC is isosceles and sets up the base angle calculation:
$\dfrac{180 - 80}{2}$ **[M1]**

50 **[A1]**

[Total Marks: 7]

21 (a) Angle $PTS = 66°$ **[B1]**

Reason: co-interior angles sum to 180° (PW is parallel to US) **[B1]**

(b) Sum of interior angles of a pentagon $= (5 - 2) \times 180° = 540°$ **[B1]**

$114 + (3x + 16) + (4x - 14) + (2x + 34) + 66 = 540$ **[M1]**

$9x + 216 = 540$ leading to $9x = 324$ and $x = 36$ **[A1]**

(c) (i) 114 **[B1]**

(ii) Angle $PQR = 3(36) + 16 = 124°$ and angle $WQR = 180° - 124° = 56°$ **[M1]**

56 **[A1]**

(d) Exterior angle $= \dfrac{180°}{12} = 15°$ **[M1]**

$n = \dfrac{360}{15}$ **[M1]**

24 **[A1]**

[Total Marks: 11]